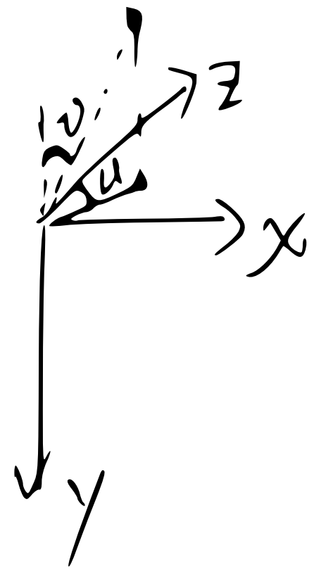


2.b spherical/portrait projection ②

2.b.3 $(x,y,z) \to (u,v)$

$$\begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow \left\|\begin{pmatrix}x\\y\\z\end{pmatrix}\right\|^{-1}\begin{pmatrix}x\\y\\z\end{pmatrix}, \quad \begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow \begin{pmatrix}y\\x\\z\end{pmatrix}$$

$$\therefore \begin{pmatrix}u\\v\end{pmatrix} = r\begin{pmatrix}\tan^{-1}\left(\frac{x}{z}\right)\\ \pi-\cos^{-1}y\end{pmatrix}$$

$$\begin{pmatrix}u\\v\end{pmatrix} \leftarrow \begin{pmatrix}-u\\v\end{pmatrix}$$

2.b.4. $\begin{pmatrix}u\\v\end{pmatrix} \to \begin{pmatrix}x\\y\\z\end{pmatrix}$

$$\begin{pmatrix}u\\v\end{pmatrix} \leftarrow \begin{pmatrix}-u\\v\end{pmatrix}, \begin{pmatrix}u\\v\end{pmatrix} \leftarrow r^{-1}\begin{pmatrix}u\\v\end{pmatrix}, v \leftarrow \pi - v$$

$$\therefore \begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}sv\,su\\ cv\\ sv\,cu\end{pmatrix} \quad \therefore \begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow \begin{pmatrix}y\\x\\z\end{pmatrix}$$

1.6

Given camera K, R, t,

$$\because \lambda\begin{pmatrix}x\\1\end{pmatrix} = K(R,t)\begin{pmatrix}X\\1\end{pmatrix} = KX_{cam}$$

K is just affine

$$\therefore \lambda = X_{cam}(3)$$

$\therefore$ if $\lambda \le 0$, X can't be projected onto image.

2. spherical projection.

2.3 $(x,y,z) \to (u,v)$

$$\begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow \left\|\begin{pmatrix}x\\y\\z\end{pmatrix}\right\|^{-1}\begin{pmatrix}x\\y\\z\end{pmatrix}$$

$$\therefore u = r\tan^{-1}(x,z) = r\tan^{-1}\left(\frac{x}{z}\right)$$

$$v = r(\pi - \cos^{-1}y)$$

2.4 $(u,v) \to \begin{pmatrix}x\\y\\z\end{pmatrix}$:

$$\begin{pmatrix}u\\v\end{pmatrix} \leftarrow r^{-1}\begin{pmatrix}u\\v\end{pmatrix}, \quad v \leftarrow \pi - v$$

$$\therefore \begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}sv\,su\\ cv\\ sv\,cu\end{pmatrix}$$

(Not check)

1. camera.

1.3 projectivity: given K, R, t

$$\Rightarrow \lambda\begin{pmatrix}x\\1\end{pmatrix} = K(R,t)\begin{pmatrix}X\\1\end{pmatrix}, \quad x \in \mathbb{R}^2, \; X \in \mathbb{R}^3$$

1.4 homography

$$\lambda_1\begin{pmatrix}x_1\\1\end{pmatrix} = K_1R_1X, \quad \lambda_2\begin{pmatrix}x_2\\1\end{pmatrix} = K_2R_2X$$

$$\therefore \lambda_2\begin{pmatrix}x_2\\1\end{pmatrix} = K_2R_2\lambda_1R_1^{-1}K_1^{-1}\begin{pmatrix}x_1\\1\end{pmatrix}$$

$$\therefore \lambda\begin{pmatrix}x_2\\1\end{pmatrix} = H_{2\to1}\begin{pmatrix}x_1\\1\end{pmatrix}$$

1.5 affine: given K, R, t for camera,

$\because$ K is always affine

$\therefore$ camera $K=I$, R affine, $t=0$ can represent image affine trans.

4.b.4 $\begin{pmatrix}u\\v\end{pmatrix} \to \begin{pmatrix}x\\y\\z\end{pmatrix}$

solve: $\begin{pmatrix}u\\v\end{pmatrix} \leftarrow \begin{pmatrix}-u\\v\end{pmatrix}$, $\begin{pmatrix}u\\v\end{pmatrix} \leftarrow r^{-1}\begin{pmatrix}u\\v\end{pmatrix}$

$\therefore \begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}u\\v\end{pmatrix} - \begin{pmatrix}t_1\\t_2\end{pmatrix}$, $z = 1 - t_3$

$\therefore \begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow \begin{pmatrix}y\\x\\z\end{pmatrix}$

4. plane projection ⑤

4.3 $\begin{pmatrix}x\\y\\z\end{pmatrix} \to \begin{pmatrix}u\\v\end{pmatrix}$

solve: $\begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow z^{-1}\begin{pmatrix}x\\y\\z\end{pmatrix}$, $\begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow (1-t_3)\begin{pmatrix}x\\y\\z\end{pmatrix}$

$\therefore z = (1-t_3)$, $\begin{pmatrix}u\\v\end{pmatrix} = r\left[\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}t_1\\t_2\end{pmatrix}\right]$

4.4 $\begin{pmatrix}u\\v\end{pmatrix} \to \begin{pmatrix}x\\y\\z\end{pmatrix}$

solve: $\begin{pmatrix}u\\v\end{pmatrix} \leftarrow r^{-1}\begin{pmatrix}u\\v\end{pmatrix}$,

$\therefore \begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}u\\v\end{pmatrix} - \begin{pmatrix}t_1\\t_2\end{pmatrix}$,

$z = 1 - t_3$

4.b. plane portrait projection

4.b.3 $\begin{pmatrix}x\\y\\z\end{pmatrix} \to \begin{pmatrix}u\\v\end{pmatrix}$.

solve: $\begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow \begin{pmatrix}y\\x\\z\end{pmatrix}$, $\begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow z^{-1}\begin{pmatrix}x\\y\\z\end{pmatrix}$

$\begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow (1-t_3)\begin{pmatrix}x\\y\\z\end{pmatrix}$, $\therefore z = 1 - t_3$

$\therefore \begin{pmatrix}u\\v\end{pmatrix} = r\left[\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}t_1\\t_2\end{pmatrix}\right]$

$\therefore \begin{pmatrix}u\\v\end{pmatrix} \leftarrow \begin{pmatrix}-u\\v\end{pmatrix}$

3. cylindrical projection.

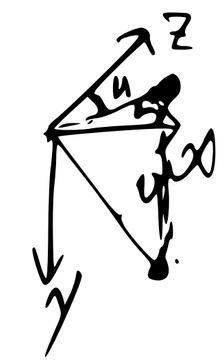
3.3 $\begin{pmatrix}x\\y\\z\end{pmatrix} \to \begin{pmatrix}u\\v\end{pmatrix}$

$\begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow (x^2+z^2)^{-\frac{1}{2}}\begin{pmatrix}x\\y\\z\end{pmatrix}$

$\therefore \begin{pmatrix}u\\v\end{pmatrix} \leftarrow r\begin{pmatrix}\tan^{-1}(x,z)\\y\end{pmatrix}$

3.4 $\begin{pmatrix}u\\v\end{pmatrix} \to \begin{pmatrix}x\\y\\z\end{pmatrix}$

$\begin{pmatrix}u\\v\end{pmatrix} \leftarrow r^{-1}\begin{pmatrix}u\\v\end{pmatrix}$, $\therefore \begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}su\\v\\cu\end{pmatrix}$



3.b cylindrical portrait projection ⑤

3.b.3 $\begin{pmatrix}x\\y\\z\end{pmatrix} \to \begin{pmatrix}u\\v\end{pmatrix}$

solve: $\begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow \begin{pmatrix}y\\x\\z\end{pmatrix}$, $\begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow (x^2+z^2)^{-\frac{1}{2}}\begin{pmatrix}x\\y\\z\end{pmatrix}$

$\begin{pmatrix}u\\v\end{pmatrix} = r\begin{pmatrix}\tan^{-1}(\frac{x}{z})\\y\end{pmatrix}$, $\begin{pmatrix}u\\v\end{pmatrix} \leftarrow \begin{pmatrix}-u\\v\end{pmatrix}$

3.b.4 $\begin{pmatrix}u\\v\end{pmatrix} \to \begin{pmatrix}x\\y\\z\end{pmatrix}$; solve: $\begin{pmatrix}u\\v\end{pmatrix} \leftarrow \begin{pmatrix}-u\\v\end{pmatrix}$, $\begin{pmatrix}u\\v\end{pmatrix} \leftarrow r^{-1}\begin{pmatrix}u\\v\end{pmatrix}$

$\therefore \begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}su\\v\\cu\end{pmatrix}$, $\therefore \begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow \begin{pmatrix}y\\x\\z\end{pmatrix}$